

Review Questions : Quantum Mechanics-1
Module-4 : SOLUTION OF SCHRÖDINGER EQUATION FOR

3D PROBLEM

Part A

The One-Dimensional Harmonic Oscillator

1. Write the potential-energy function for the one-dimensional harmonic oscillator. State the energy eigenvalues of the quantum harmonic oscillator. What is meant by the ground state of the harmonic oscillator? What is meant by an excited state?
2. Write the normalized ground-state wave function of the harmonic oscillator. Write the normalized first-excited-state wave function in position space.
3. State the parity of the ground state and first excited state. How many nodes does the ground-state wave function have? How many nodes does the first-excited-state wave function have?
4. State the relation between the energy-level spacing of a harmonic oscillator and $\hbar\omega$. What is the zero-point energy of a harmonic oscillator? Why can the energy of a quantum harmonic oscillator never be zero?
5. Starting from the requirement that the annihilation operator destroys the ground state ($\hat{a}|0\rangle=0$), write down the differential equation for $\psi_0(x)$ in position space and state its normalized solution.
6. Derive the normalized ground state spatial wavefunction $\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\left(\frac{m\omega}{2\hbar}\right)x^2}$ using $\hat{a}|0\rangle=0$. Apply the creation operator $\hat{a}^\dagger$ to $\psi_0(x)$ to obtain the normalized spatial wavefunction $\psi_1(x)$ for the first excited state.

Matrix Representation of Harmonic-Oscillator Operators

13. Define the ladder operators a and $a^\dagger$. State the commutation relation between a and $a^\dagger$.
14. Express the position operator $\hat{X}$ in terms of a and $a^\dagger$. Express the momentum operator $\hat{P}$ in terms of a and $a^\dagger$.
15. State the action of the lowering operator and the raising operator on $|n\rangle$.
16. Write the Hamiltonian of the harmonic oscillator in terms of $a^\dagger a$.
17. What is the matrix representations of the annihilation operator and the creation operator in the energy eigen basis? State the matrix elements $\langle m | \hat{X} | n \rangle$ qualitatively.
18. Truncating the Hilbert space to the subspace spanned by the first three energy eigenstates $|0\rangle, |1\rangle$ and $|2\rangle$ of a 1D harmonic oscillator:
 - Construct the 3×3 matrix representations for $\hat{a}$ and $\hat{a}^\dagger$.
 - Construct the corresponding 3×3 matrix representation for the position operator $\hat{X}$.

Expectation Values

23. What are the values of $\langle \hat{X} \rangle$ and $\langle \hat{P} \rangle$ in any energy eigenstate of a one-dimensional harmonic oscillator?
24. State the relationship between $\langle \hat{X}^2 \rangle$, $\langle \hat{P}^2 \rangle$, and the energy of a harmonic oscillator.

Separation of Variables in Three Dimensions

28. State the three-dimensional time-independent Schrödinger equation in Cartesian coordinates.
29. What is meant by separation of variables?
30. Write the product form assumed for a separable three-dimensional wave function.

31. What condition must the potential satisfy for Cartesian separation of variables to be possible? If $V(x, y, z) = V_x(x) + V_y(y) + V_z(z)$, how is the total energy related to the energies in the three coordinate directions?
32. What are the quantum numbers associated with a separable Cartesian problem? Write the general form of a separable three-dimensional eigenfunction.

Three-Dimensional Box Potential

35. Define a rectangular infinite potential box. What is the potential inside and outside a three-dimensional infinite box?
36. Write the normalized energy eigenfunction for a rectangular box. Write the energy eigenvalues for a rectangular box with side lengths L_x, L_y, L_z . What are the allowed values of the quantum numbers n_x, n_y, n_z ?
37. What is meant by degeneracy of an energy level? Explain why a cubic box has more degeneracy than a rectangular box with unequal side lengths.
38. Give one example of two distinct states having the same energy in a cubic box. Is the ground state of a three-dimensional cubic box degenerate? Explain briefly.

Three-Dimensional Harmonic Oscillator

45. Write the potential-energy function for an anisotropic three-dimensional harmonic oscillator. State the energy eigenvalues of an anisotropic three-dimensional oscillator.
46. What is meant by an isotropic harmonic oscillator? State the energy eigenvalues of an isotropic three-dimensional oscillator.
47. What are the quantum numbers of the three-dimensional harmonic oscillator?

48. State the ground-state energy of the isotropic three-dimensional oscillator. Is the ground state of the isotropic oscillator degenerate?
49. What condition on n_x, n_y, n_z gives states with the same energy in an isotropic oscillator? State the degeneracy of the energy level characterized by $N = n_x + n_y + n_z$. Why is an anisotropic oscillator generally less degenerate than an isotropic oscillator?

Part B

The One-Dimensional Harmonic Oscillator

1. Explain the physical meaning of the energy eigenvalues of the quantum harmonic oscillator.
2. Write the ground-state and first-excited-state wave functions of the harmonic oscillator. Sketch their qualitative forms and indicate their nodes. Show that the first-excited-state wave function is orthogonal to the ground-state wave function.
3. Explain the parity properties of the ground state and first excited state.
4. Show that the ground-state energy of the harmonic oscillator is $\hbar \omega / 2$, not zero.
5. Compare the probability densities of the ground state and first excited state.

Matrix Representation and Expectation Values

7. Using the energy eigen basis, write the first few rows and columns of the matrix representation of the annihilation operator $\hat{a}$.
8. Derive the matrix representation of the position operator using $\hat{X} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^\dagger)$
9. Derive the matrix representation of the momentum operator using

$$\hat{P} = i\sqrt{\frac{m\hbar\omega}{2}}(\hat{a} - \hat{a}^\dagger)$$

10. Show that $\langle n | \hat{X} | n \rangle = 0$, for any energy eigenstate of the harmonic oscillator.
11. Calculate $\langle \hat{X}^2 \rangle$ and $\langle \hat{P}^2 \rangle$ for the ground state and verify the uncertainty relation.
12. Show that the expectation value of the Hamiltonian in the n th energy eigen state is

$$\langle \hat{H} \rangle = \hbar \omega \left(n + \frac{1}{2} \right).$$

13. For the harmonic oscillator state $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, calculate $\langle \hat{X} \rangle$.

Separation of Variables

15. Starting with the three-dimensional Schrödinger equation, assume

$$\psi(x, y, z) = X(x)Y(y)Z(z).$$

Show how the equation separates when $V(x, y, z) = V_x(x) + V_y(y) + V_z(z)$. Explain the physical meaning of the three separation constants in a Cartesian problem. Show that the total energy is the sum of the energies associated with the x , y , and z motions.

16. Write the general form of the wave function and energy for a three-dimensional separable system.

Rectangular and Cubic Box Potentials

19. Derive the allowed energy eigenvalues for a particle in a rectangular box of dimensions L_x, L_y, L_z . Write the normalized eigenfunction for a particle in a rectangular box and verify its normalization.
20. Find the ground-state energy of a particle in a cubic box of side L . Find the first three distinct energy levels of a particle in a cubic box and state their degeneracies.

21. Determine the degeneracy of the energy corresponding to $n_x^2 + n_y^2 + n_z^2 = 14$.

Explain why the states $(1, 1, 2)$, $(1, 2, 1)$, and $(2, 1, 1)$ have the same energy in a cubic box.

22. Compare the energy spectrum of a rectangular box with unequal side lengths and a cubic box. Show that the ground state of a rectangular or cubic box is nondegenerate.

23. For a cubic box, determine whether the states $(1, 2, 3)$ and $(3, 2, 1)$ are degenerate. Explain.

Three-Dimensional Harmonic Oscillator

28. Derive the energy eigenvalues of an anisotropic three-dimensional harmonic oscillator from the corresponding one-dimensional energies.

29. Write the eigenfunctions of the three-dimensional harmonic oscillator as products of one-dimensional oscillator eigenfunctions.

30. Find the ground-state energy of the anisotropic oscillator with frequencies $\omega_x, \omega_y, \omega_z$.

31. For an isotropic oscillator, find the energy and degeneracy of the level with

$$n_x + n_y + n_z = 2.$$

32. List all the states belonging to the $N = 2$ energy level of the isotropic oscillator.

33. Find the degeneracy of the $N = 3$ energy level of the isotropic oscillator.

34. Explain why the isotropic oscillator has greater degeneracy than the anisotropic oscillator.

35. Compare the first two energy levels of a one-dimensional oscillator and a three-dimensional isotropic oscillator.

Part C

The One-Dimensional Harmonic Oscillator

1. Derive the energy eigenvalues of the one-dimensional quantum harmonic oscillator using ladder operators. Explain the significance of the zero-point energy.
2. Discuss the ground state and first excited state of the harmonic oscillator in position space. Derive their normalized wave functions, sketch them, and explain their parity and node structure.
3. Develop the matrix representation of the harmonic-oscillator operators $\hat{a}$, $\hat{a}^\dagger$, $\hat{X}$, $\hat{P}$, and $\hat{H}$ in the energy eigenbasis.
4. Derive the expectation values $\langle \hat{X} \rangle$, $\langle \hat{P} \rangle$, $\langle \hat{X}^2 \rangle$, and $\langle \hat{P}^2 \rangle$ for the n th stationary state of the harmonic oscillator. Calculate the product $(\Delta x \Delta p)$ and hence obtain the uncertainty principle.
5. Explain how ladder operators connect the different energy eigenstates of the harmonic oscillator. Discuss their action on the ground state and excited states.

Three-Dimensional Separation of Variables

6. Starting from the three-dimensional time-independent Schrödinger equation, derive the separation-of-variables method for a potential of the form

$$V(x, y, z) = V_x(x) + V_y(y) + V_z(z).$$

Explain the meaning of the separation constants and the resulting total energy.

7. Discuss the general treatment of three-dimensional problems in Cartesian coordinates. Explain the role of product wave functions, independent quantum numbers, and additive energies.

Rectangular and Cubic Box Potentials

8. Solve the Schrödinger equation for a particle in a rectangular three-dimensional infinite potential box. Derive the normalized eigenfunctions and energy eigenvalues.
9. Discuss degeneracy in a three-dimensional rectangular box and a cubic box. Explain why degeneracy occurs and compare the energy spectra for unequal and equal side lengths.
10. For a particle in a cubic box, derive the first few energy levels and their degeneracies. Explain the physical meaning of degenerate eigenstates.
11. Derive the energy eigenvalues and eigenfunctions for the cubic box and discuss the states corresponding to the first excited energy level.
12. Explain how the symmetry of the cubic box leads to degeneracy. Give suitable examples of distinct quantum states with equal energy.

Three-Dimensional Harmonic Oscillator

13. Derive the energy eigenvalues and eigenfunctions of the anisotropic three-dimensional harmonic oscillator using separation of variables.
14. Explain the difference between anisotropic and isotropic three-dimensional harmonic oscillators. Discuss their energy spectra and degeneracies.
15. Derive the energy spectrum of the isotropic three-dimensional oscillator:

$$E_N = \hbar \omega \left(N + \frac{3}{2} \right), \quad N = n_x + n_y + n_z$$

Derive the degeneracy formula $g_N = \frac{(N+1)(N+2)}{2}$ for the isotropic three-dimensional harmonic oscillator.

16. List and discuss the states corresponding to the ground, first excited, and second excited levels of the isotropic three-dimensional oscillator.

17. Compare the degeneracy of the isotropic three-dimensional oscillator with that of the three-dimensional cubic box. Explain the origin of degeneracy in each system.
18. Discuss the physical interpretation of the quantum numbers n_x, n_y, n_z in the three-dimensional harmonic oscillator.
19. Explain how the three-dimensional harmonic oscillator can be viewed as three independent one-dimensional oscillators. Derive the total energy and discuss the role of separability.